

Basic Formulas

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| 1. =SUM (A1:A10)
<i>Example: =SUM (10,20) → 30</i> | Adds numbers in a range. |
| 2. =AVERAGE (A1:A10)
<i>Example: =AVERAGE (10,20,30) → 20</i> | Returns the mean. |
| 3. =MEDIAN (A1:A10)
<i>Example: =MEDIAN (10,20,30,40) → 25</i> | Returns the middle value. |
| 4. =MODE (A1:A10)
<i>Example: =MODE (1,2,2,3) → 2</i> | Most frequent value. |
| 5. =MIN (A1:A10)
<i>Example: =MIN (5,7,9) → 5</i> | Smallest value in range. |
| 6. =MAX (A1:A10)
<i>Example: =MAX (5,7,9) → 9</i> | Largest value in range. |
| 7. =ROUND(A1,2)
<i>Example: =ROUND (12.345,2) → 12.35</i> | Rounds to 2 decimal places. |
| 8. =ROUNDUP(A1,0)
<i>Example: =ROUNDUP (12.1,0) → 13</i> | Always rounds up. |
| 9. =ROUNDDOWN(A1,0)
<i>Example: =ROUNDDOWN (12.9,0) → 12</i> | Always rounds down. |
| 10. =TRUNC(A1,2)
<i>Example: =TRUNC (12.345,2) → 12.34</i> | Cuts decimals without rounding. |

Logical Functions

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| 11. =IF(A1>50,"Pass","Fail") | Returns values based on condition. |
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| 12. =AND(A1>50, B1<100)
<i>Example: Both must be TRUE → TRUE/FALSE</i> | Checks multiple conditions. |
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| 13. =OR(A1>50, B1<100) | Any condition TRUE returns TRUE. |
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| 14. =NOT(A1=10) | Returns TRUE if condition is FALSE. |
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| 15. =IFERROR(A1/B1,"Error") | Handles division by zero. |
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Conditional Aggregation

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| 16. =COUNT (A1:A10)
<i>Example: =COUNT(1,2,"x") → 2</i> | Counts numbers. |
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| 17. =COUNTA (A1:A10)
<i>Example: =COUNTA(1,2,"x","") → 3</i> | Counts non-empty. |
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| 18. =COUNTBLANK (A1:A10)
<i>Example: =COUNTBLANK({"A","","B"}) → 1</i> | Counts blanks. |
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| 19. =COUNTIF (A1:A10,">10")
<i>Example: =COUNTIF ({5,15,20},">10") → 2</i> | Counts >10. |
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| 20. =COUNTIFS (A1:A10,">10", B1:B10,"Yes")
criteria. | Counts with multiple criteria. |
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| 21. =SUMIF (A1:A10,">10")
<i>Example: =SUMIF ({5,15,20},">10") → 35</i> | Sum if >10. |
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| 22. =SUMIFS (A1:A10, B1:B10,"Yes") | Sum with multiple criteria. |
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23. =AVERAGEIF (A1:A10, ">10")

Avg of values >10.

24. =AVERAGEIFS (A1:A10, B1:B10, "Yes")

Avg with multiple criteria.

25. =IF(A1>10, "Pass", "Fail")

Conditional output

Lookup & Reference

26. = VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Looks for a value in the first column of a range and returns a value from another column in the same row.

- **Example:**

=VLOOKUP(101, A2:D10, 3, FALSE) → Searches for 101 in column A and returns the value from 3rd column in that row.

27. =HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

- Works like VLOOKUP but searches horizontally in the first row.

- **Example:**

=HLOOKUP(50, A1:H5, 3, FALSE) → Searches for 50 in row 1 and returns value from 3rd row in that column.

28. =INDEX(array, row_num, [col_num])

- Returns the value of a cell from a given row and column in a range.

- **Example:**

=INDEX(A2:C10, 2, 3) → Returns the value from row 2, column 3 of the range.

29. =MATCH(lookup_value, lookup_array, [match_type])

- Returns the position of a value in a range.

- **Example:**

=MATCH(45, A1:A10, 0) → Finds position of 45 in the range

30. =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

- Modern replacement for VLOOKUP/HLOOKUP, more flexible.
 - **Example:**
=XLOOKUP(45, A1:A10, B1:B10, "Not Found") → If 45 is found in A column, returns corresponding value from column B.
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31. =OFFSET(reference, rows, cols, [height], [width])

- Returns a cell reference that is offset from a starting cell by rows and columns.
 - **Example:**
=OFFSET(A1,2,3) → Returns the cell that is 2 rows down and 3 columns right of A1.
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32. =CHOOSE(index_num, value1, [value2], ...)

- Chooses a value from a list by position.
 - **Example:**
=CHOOSE(2, "Red", "Blue", "Green") → Returns "Blue".
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Reference

33. Relative Reference → =A1+B1 Cell references change when copied.

- **Example: Copying from row 1 to row 2 becomes =A2+B2.**
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34. Absolute Reference → =\$A\$1+\$B\$1

Cell reference does not change when copied.

- **Example: Always adds values from A1 and B1.**
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35. Mixed Reference → =A\$1+\$B1

Locks either row or column.

Example: A\$1 locks row 1, \$B1 locks column B.

Text Functions

36. =LEN(A1) Counts characters (including spaces).

- =LEN(text)
 - Example: =LEN("Excel") → 5
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37. =LEFT(A1,4) Returns first N characters.

- =LEFT(text, num_chars)
 - Example: =LEFT("Excel",4) → "Exce"
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38. =RIGHT(A1,3) Returns last N characters.

- =RIGHT(text, num_chars)
 - Example: =RIGHT("Excel",3) → "cel"
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39. =MID(A1,2,3) Extracts substring starting at a position.

- =MID(text, start_num, num_chars)
 - Example: =MID("Excel",2,3) → "xce"
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40. =CONCAT(A1,B1) Joins text values.

- =CONCAT(text1, [text2], ...)
 - Example: =CONCAT("Data","Science") → "DataScience"
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41. =TEXTJOIN("-",TRUE,A1:A3) Joins text with delimiter.

- =TEXTJOIN(delimiter, ignore_empty, text1, ...)
 - Example: =TEXTJOIN("-",TRUE,{"Red","Blue","Green"}) → "Red-Blue-Green"
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42. =TRIM(A1) Removes extra spaces.

- **=TRIM(text)**
 - **Example:** =TRIM(" Excel Guide ") → "Excel Guide"
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43. =PROPER(A1) Converts to Proper Case.

- **=PROPER(text)**
 - **Example:** =PROPER("data analysis") → "Data Analysis"
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44. =UPPER(A1) Converts to UPPERCASE.

- **Example:** =UPPER("excel") → "EXCEL"
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45. =LOWER(A1) Converts to lowercase.

- **Example:** =LOWER("EXCEL") → "excel"
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46. =REPLACE(A1,2,3,"XYZ") Replaces part of text.

- **=REPLACE(old_text, start_num, num_chars, new_text)**
 - **Example:** =REPLACE("Excel",2,3,"XYZ") → "EXYZI"
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47. =SUBSTITUTE(A1,"old","new") Replaces specific text.

- **=SUBSTITUTE(text, old_text, new_text, [instance_num])**
 - **Example:** =SUBSTITUTE("Good morning","morning","night") → "Good night"
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48. =FIND("x",A1) Returns position of text (case-sensitive).

- **=FIND(find_text, within_text)**
 - **Example:** =FIND("x","Excel") → 2
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49. =SEARCH("x",A1) Same as FIND but not case-sensitive.

- **Example:** =SEARCH("X","Excel") → 2

Date & Time

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- 50.=TODAY()** Returns current date.
Example: =TODAY() → 14-09-2025
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- 51.=NOW()** Returns current date & time.
Example: =NOW() → 14-09-2025 21:07
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- 52.=DAY(A1)** Extracts day from date.
Example: If A1 = 14-09-2025, =DAY(A1) → 14
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- 53.=MONTH(A1)** Extracts month from date.
Example: If A1 = 14-09-2025, =MONTH(A1) → 9
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- 54.=YEAR(A1)** Extracts year from date.
Example: If A1 = 14-09-2025, =YEAR(A1) → 2025
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- 55.=WEEKDAY(A1)** Returns weekday number (1=Sunday, 2=Monday ...).
Example: If A1 = 14-09-2025 (Sunday), =WEEKDAY(A1) → 1
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- 56.=EDATE(A1,6)** Returns date after 6 months.
Example: If A1 = 14-09-2025, =EDATE(A1,6) → 14-03-2026
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- 57.=DATEDIF(A1,B1,"Y")** Returns age in years.
Example: If A1 = 14-09-2000, B1 = 14-09-2025, → 25
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- 58.=NETWORKDAYS(A1,B1)** Working days between two dates.
Example: If A1 = 01-09-2025, B1 = 14-09-2025, → 10
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Statistical & Math Functions

59. **=STDEV(A1:A10)** Standard deviation of values.
Example: If A1:A10 = {2,4,4,4,5,5,7,9}, → 2
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60. **=VAR(A1:A10)** Variance of values.
Example: If A1:A10 = {2,4,4,4,5,5,7,9}, → 4
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61. **=CORREL(A1:A10,B1:B10)** Correlation coefficient.
Example: If A1:A5 = {1,2,3,4,5}, B1:B5 = {2,4,6,8,10}, → 1
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62. **=COVAR(A1:A10,B1:B10)** Covariance.
Example: If A1:A5 = {1,2,3}, B1:B5 = {2,4,6}, → 2.33
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63. **=FREQUENCY(A1:A10,B1:B5)** Frequency distribution.
Example: A1:A10 = {5,7,3,8,5,10,12,7}, B1:B5 = {5,10,15} → {2,3,3}
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64. **=PERCENTILE(A1:A10,0.9)** 90th percentile.
Example: If A1:A10 = {1,2,3,4,5,6,7,8,9,10}, → 9
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65. **=RANK(A1,A1:A10)** Rank of a number.
Example: If A1=85 and A1:A10={90,85,70,60}, → 2
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66. **=QUARTILE(A1:A10,1)** First quartile.
Example: If A1:A10={10,20,30,40,50,60,70,80}, → 27.5
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67. **=ABS(A1)** Absolute value.
Example: If A1 = -15, =ABS(A1) → 15
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68. **=POWER(A1,2)** Square/power.
Example: If A1=5, =POWER(A1,2) → 25
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69. **=SQRT(A1)** Square root.
Example: If A1=49, =SQRT(A1) → 7
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70. **=PI()** Returns 3.14159.
Example: =PI() → 3.14159
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71. =RAND()

Random number (0–1).

Example: =RAND() → 0.6732 (varies)

72. =RANDBETWEEN(1,100)

Random integer.

Example: =RANDBETWEEN(1,100) → 57 (varies)

Data Analysis Helpers

73. =UNIQUE(A1:A10)

Returns unique values from a range.

Example: A1:A10 = {A,B,B,C,A} → {A,B,C}

74. =SORT(A1:A10,1,TRUE)

Sorts values ascending.

Example: A1:A5={30,10,50,20} → {10,20,30,50}

75. =FILTER(A1:C10,B1:B10="Yes")

Filters rows where column B = Yes.

Example: If Column B has Yes/No, only rows with Yes return.

76. =TRANSPOSE(A1:B2)

Converts rows to columns (and vice versa).

Example: If A1:B2 =

1 2
3 4

Become:-

1 3
2 4
